A Partial Literature Answer to the Frechet Smooth + LUR Renorming Question

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Status. This is a lightweight literature-implied status note:

`literature_implied_answer (partial subcases).`

It records known positive subcases of a question in Gothwal–Shekhar, *(Asymptotic) uniform smoothness, ball separation and residuality results*, arXiv:2502.18217.

Source question. In Section 5, "Two open problems", the source asks:

If X has an equivalent Frechet smooth norm and an equivalent LUR norm,

does X admit an equivalent norm which is both Frechet smooth and LUR?

The question is presented as a consequence that would follow from a positive answer to the preceding residuality question for Frechet smooth norms.

Supporting literature. Hajek–Prochazka, *C^k-smooth approximations of LUR norms*, arXiv:0901.3623, gives several positive cases. In the introduction they recall that every separable Asplund space admits an equivalent norm which is simultaneously C^1-Frechet smooth and LUR. Since the existence of a Frechet smooth norm gives a C^1-smooth bump and hence Asplundness, this covers the separable case of the source question.

More generally, their Corollary 4.4 states that if X admits a C^k-Frechet smooth norm and X is Vasak (WCD), WLD, or $C(K)$ with K Valdivia compact, then X admits an equivalent C^1-Frechet smooth LUR norm. Taking $k = 1$ gives a direct positive answer to the source question on these classes.

Scope limitations. The supporting paper predates arXiv:2502.18217 and is not presented there as an answer to that paper's question; the implication is an agent-identified literature match. The full arbitrary Banach-space question remains open by this packet. The earlier residuality question,

are Frechet smooth equivalent norms residual whenever one exists?

also remains open; Hajek–Prochazka explicitly note that density/residuality of C^1-Frechet smooth norms is unknown even for some $C([0, \alpha])$ spaces.

References.

- D. Gothwal and V. Shekhar, *(Asymptotic) uniform smoothness, ball separation and residuality results*, arXiv:2502.18217.
- P. Hajek and A. Prochazka, *C^k-smooth approximations of LUR norms*, arXiv:0901.3623.